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**MPC-006**

**M.A. IN PSYCHOLOGY (MAPC)**

**Term-End Examination**

**December, 2013**

**MPC-006 : STATISTICS IN PSYCHOLOGY**

*Time : 2 hours*

*Maximum Marks : 50*

**Note :** Answer *any five* questions. Each question carries **10** marks. Only simple calculator is *permitted*.

1. Define statistics and differentiate between descriptive and inferential statistics. **2+8**
2. What do you mean by decision errors ? Discuss applications of one-tailed and two - tailed hypothesis tests in statistics. **6+4**
3. From the following data, find Rank-Difference Coefficient of correlation : **10**

| Student | Score on<br>Test I | Score on<br>Test II |
|---------|--------------------|---------------------|
|         | X                  | Y                   |
| A       | 10                 | 16                  |
| B       | 15                 | 16                  |
| C       | 11                 | 24                  |
| D       | 14                 | 18                  |
| E       | 16                 | 22                  |
| F       | 20                 | 24                  |
| G       | 10                 | 14                  |
| H       | 8                  | 10                  |
| I       | 7                  | 12                  |
| J       | 9                  | 14                  |
| N=10    |                    |                     |

4. Define regression ? Differentiate between linear 2+8  
and multiple regression by citing example.
5. Discuss the level of measurement with suitable 10  
examples.
6. What do you mean by non-parametric 2+8  
statistics ? Discuss advantages and disadvantages  
of non-parametric statistics.
7. What do you mean by two sample tests ? Write 10  
step by step procedure for Wilcoxon test for small  
sample.
8. What are assumptions of Analysis of Variance ? 4+6  
Discuss uses and limitations of ANOVA.
9. Write short notes on *any two* of the following : 10
  - (a) Type I Error
  - (b) Level of significance
  - (c) Alternative hypothesis

10. Calculate simple regression from the following raw scores. and set up regression for predicting Y from X, and also X from Y. 10

| X  | Y  | $X^2$  | $Y^2$ | XY  |
|----|----|--------|-------|-----|
| 10 | 12 | 100    | 144   | 120 |
| 11 | 18 | 121    | 324   | 198 |
| 12 | 20 | 144    | 400   | 240 |
| 9  | 10 | 81     | 100   | 90  |
| 8  | 10 | 64     | 100   | 80  |
| 50 | 70 | 510    | 1068  | 728 |
| X  | Y  | $X^2$  | $Y^2$ | XY  |
|    |    | N = 5. |       |     |